

Time-Controlled Switch Using PIC16F72



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A time-controlled switch is an automatic timer switch that turns an appliance 'on' for the desired time duration. After the preset time duration, the timer automatically switches off, disconnecting the appliance from the power supply. The time duration for which the appliance should be 'on' can be set

from 1 to 99 minutes.

This switch obviates the need to continuously monitor the appliance—an advantage over the manual switch. It can be used to switch on or switch off any electrical home appliance at a predetermined time. Switching an appliance on or off in a timely manner increases the life of the appliance and also saves power consumption.

The switch also finds industrial ap-

plications, where the machines which control the processes can be run for the desired time.

Circuit description

Fig. 1 shows the circuit of the time-controlled switch using PIC16F72 microcontroller. It comprises microcontroller PIC16F72 (IC1), regulator 7805 (IC2), two 7-segment displays (LTS542) and a few discrete components.

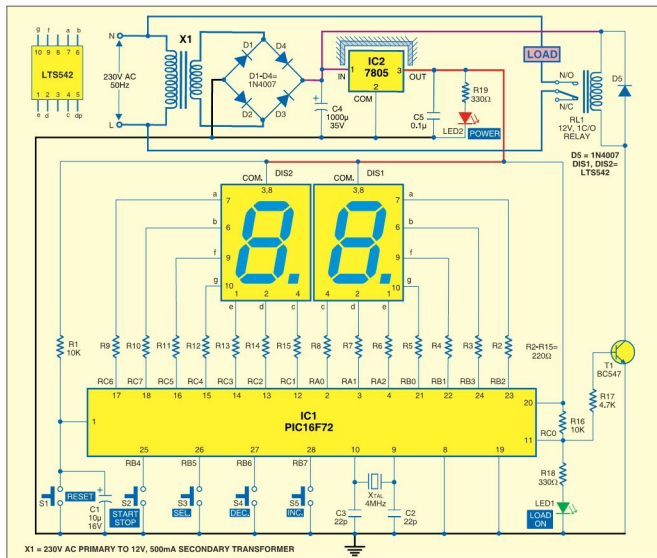


Fig. 1: Circuit of time-controlled switch using PIC16F72 microcontroller